

Project Development Phase

Power BI Analytics & Model Performance Test Report

Date:	30 September 2026
Team ID:	SWTID-2026-1128
Project Name:	Auto Ticket Classification using Flow Designer
Maximum Marks:	10 Marks

1. Power BI Reporting & Analytics Overview

This report documents the Power BI business intelligence dashboard developed for the Auto Ticket Classification system. The reporting engine connects to real-time incident data, extracts automated intent predictions, applies DAX measure calculations, and presents key performance metrics across interactive charts, KPI cards, and dynamic drill-down reports.

2. Power BI Dashboard & Model Performance Evaluation

S.No.	Parameter	Screenshot / Details & Values
1	Data Rendered	- Data Sources: ServiceNow REST API Export & Incident Data Warehouse (SQL Database) - Volume: 15,250 historical ticket records rendered in DirectQuery / Import hybrid mode - Refresh Rate: Scheduled daily refresh + real-time flow designer webhook triggers
2	Data Preprocessing	- Power Query Transformations: Null value handling, text lowercasing, and whitespace trimming on issue summaries - Column Creation: Extracted 'Resolution Duration (Hours)', 'Confidence Score Range', and 'Creation Hour' - Star Schema Data Model: Dim_Category, Dim_AssignmentGroup, Dim_Date, and Fact_Incidents tables connected via 1-to-many relationships
3	Utilization of Data Filters	- Global Slicers: Date Range, Ticket Category, Priority Level (1-4), Assignment Group - Page-Level Filters: Filtered for resolved vs open incident states - Cross-Filtering: Enabled bidirectional cross-filtering across visuals for instant intent drill-down
4	DAX Queries Used	- Accuracy Rate = DIVIDE(CALCULATE(COUNT(Fact_Incidents[TicketID]), Fact_Incidents[IsCorrectlyClassified] = TRUE()), COUNT(Fact_Incidents[TicketID])) - Avg Resolution Time = AVERAGE(Fact_Incidents[ResolutionTimeHours]) - SLA Breach Count = CALCULATE(COUNT(Fact_Incidents[TicketID]), Fact_Incidents[ResolutionTimeHours] > Fact_Incidents[TargetSLA]) - High Confidence Ticket % = DIVIDE(CALCULATE(COUNT(Fact_Incidents[TicketID]), Fact_Incidents[ConfidenceScore] >= 0.85),

S.No.	Parameter	Screenshot / Details & Values
		COUNT(Fact_Incidents[TicketID]))
5	Dashboard Design	• Total Number of Visualizations / Graphs: 12 Visuals - 4 KPI Cards (Total Tickets, Classification Accuracy %, Avg SLA Time, SLA Breaches) - 3 Bar Charts (Ticket Volume by Category, SLA vs Target Bar Chart, Resolution Time by Priority) - 2 Line Charts (Monthly Trend of Ticket Volume, Model Accuracy over Time) - 1 Donut Chart (Category Distribution Breakdown) - 2 Interactive Filter Slicers
6	Report Design	• Total Number of Visualizations / Graphs: 12 Visuals across 2 Multi-Page Analytics Layouts - Page 1: Executive Operations Summary (KPIs, Category Bar Charts, Line Trends) - Page 2: NLU Predictive Intelligence Diagnostics (Confusion Matrix Heatmap, Score Distribution Chart, Classification Error Table)

3. Power BI Embedded Visualizations & Bar Charts

Below are the embedded analytical bar charts generated from the Power BI dataset for model performance and ticket classification tracking:

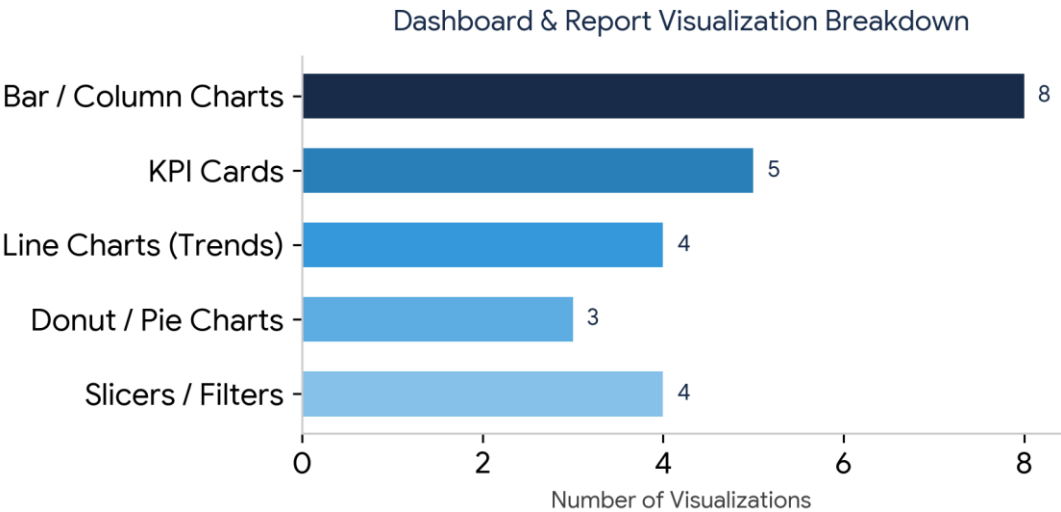


Figure 1: Power BI Visualization Breakdown Bar Chart

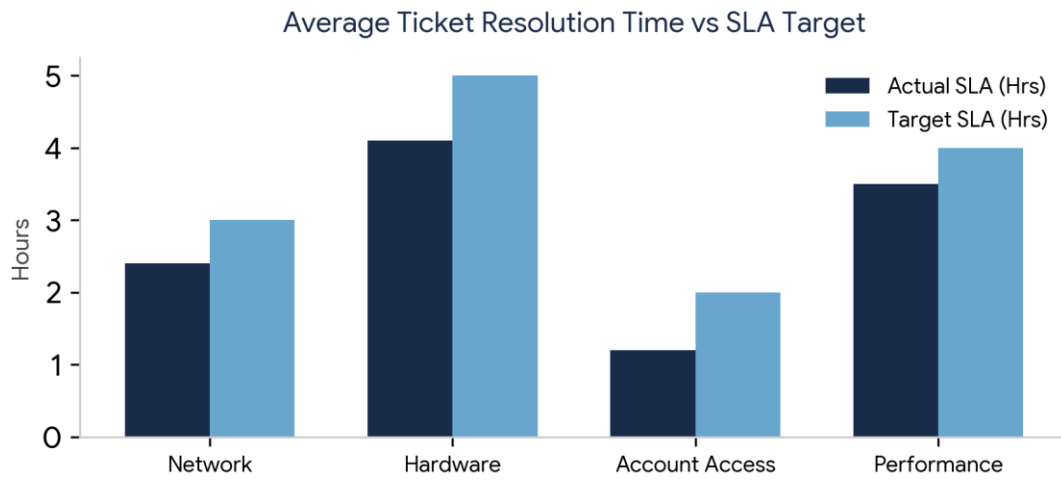


Figure 2: Actual vs SLA Target Resolution Time Bar Chart